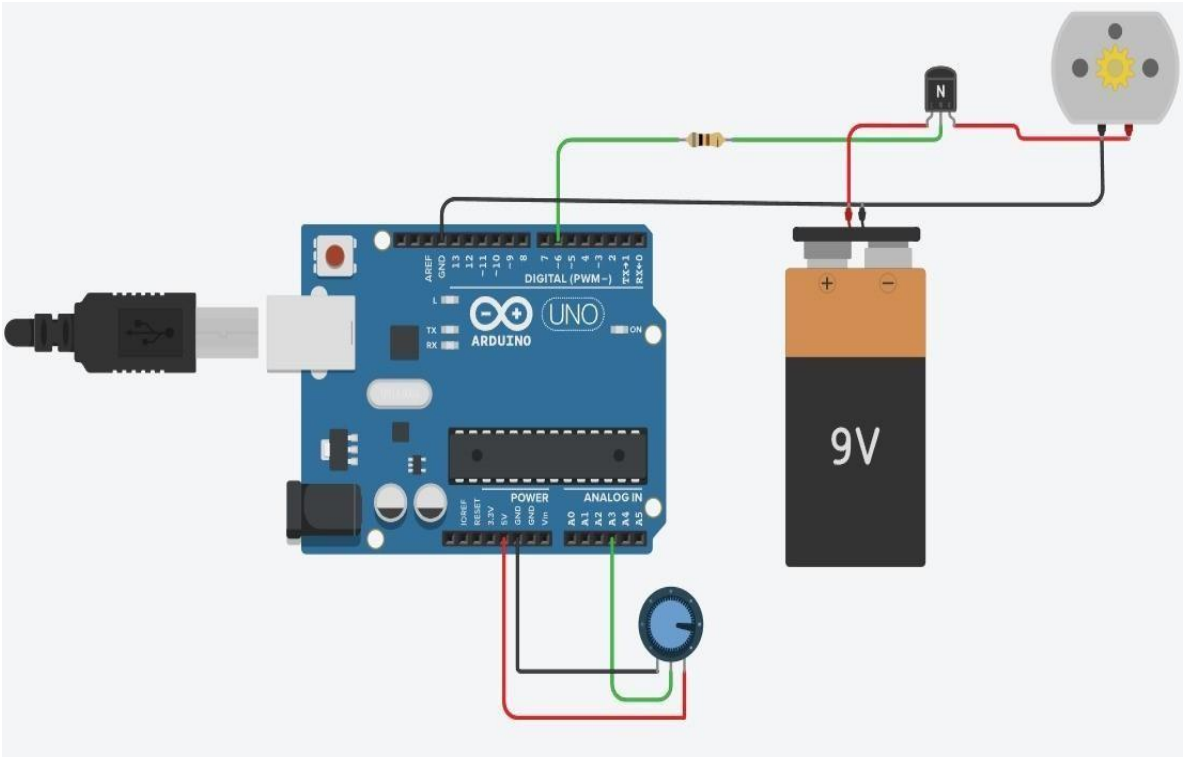
Experiment 4: Develop a program to control a DC motor with Arduino board

AIM: To develop a program to control a DC motor using an Arduino board.

COMPONENTS:

SL.NO	Name	Quantity
1	Arduino Uno	1
2	Jumper Cable	6
3	Resistance(800 Ω)	1
4	DC Motor	1
5	NPN transistor(BJT)	1
6	9V Battery	1
7	250 kΩ Potentiometer	1

CIRCUIT DIAGRAM:



SETUP:

a) Connect the circuit as per circuit.

b) Make sure VCC and Ground pins are connected properly to avoid any damage to Arduino board.

c) Open Arduino IDE then go to Tools and select appropriate Arduino board.

d) Select Tools then select the Port and select the COM port to which the board is connected.

e) Type sketch (Program) and upload to board.

CODE:

```
const int poten = A3;
int var;
void setup()
{
  pinMode(6, OUTPUT);
}
void loop()
{
  var = analogRead(poten);
  int pwmValue = map(var, 0, 1023, 0, 255);
  analogWrite(6, pwmValue);
}
```

RESULT: Successfully demonstrated control of a DC motor with Arduino board.

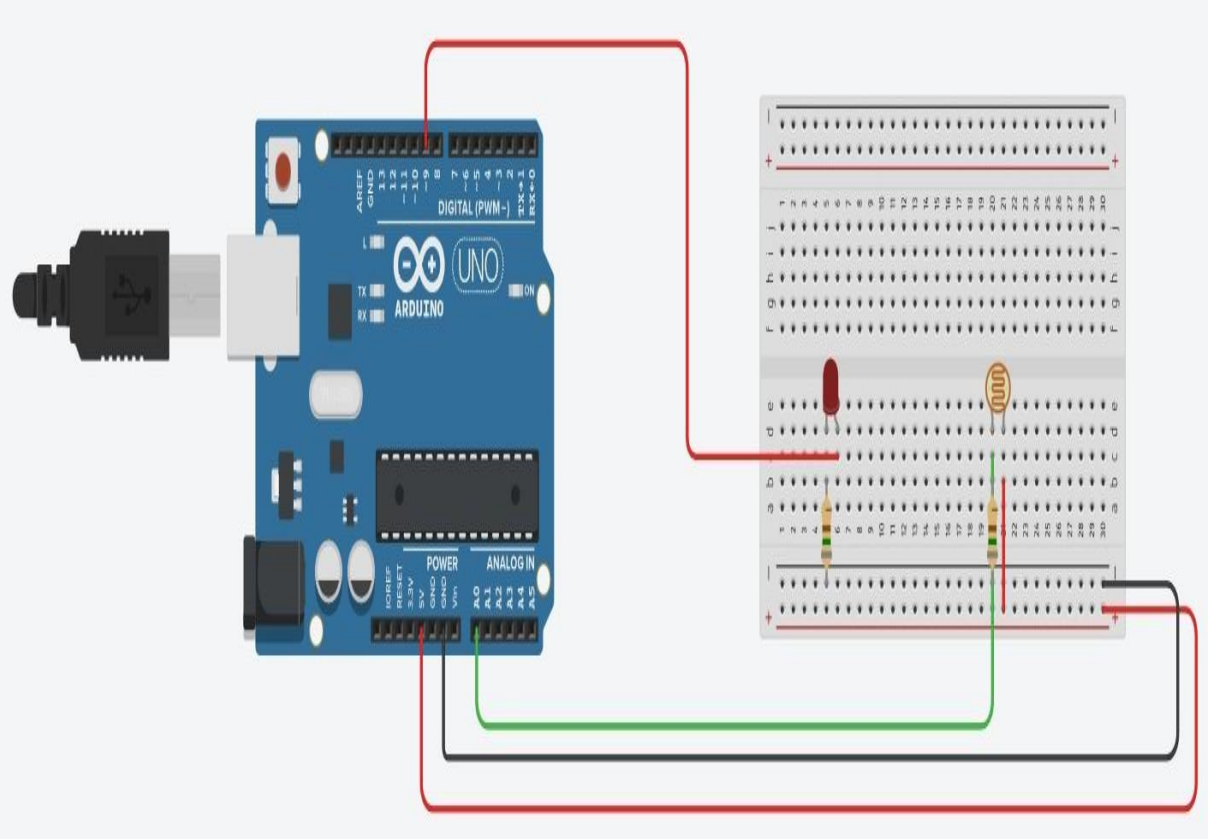
Experiment 5. Develop a program to deploy smart street light system using LDR sensor.

AIM: Develop a program to deploy smart street light system using LDR sensor.

COMPONENTS:

SL.NO	Name	Quantity
1	Arduino Uno	**1**
2	Jumper Cable	**4**
3	Bread Board	**1**
4	LED	**1**
5	Resistance (800 Ω)	**2**
6	LDR	**1**

CIRCUIT DIAGRAM:



SETUP:

a) Connect the circuit as per circuit.

b) Make sure VCC and Ground pins are connected properly to avoid any damage to Arduino board.

c) Open Arduino IDE then go to Tools and select appropriate Arduino board.

d) Select Tools then select the Port and select the COM port to which the board is connected.

e) Type sketch (Program) and upload to board.

CODE:

```
int sensorPin = A0;
int sensorValue = 0;
int led = 9;
void setup()
{
  pinMode(led, OUTPUT);
  Serial.begin(9600);
}
void loop()
 {
  sensorValue = analogRead(sensorPin);
  Serial.println(sensorValue);
  if(sensorValue < 100)
  {
    Serial.println("LED light on");
    digitalWrite(led, HIGH);
    delay(1000);
    digitalWrite(led, LOW);
  }
```

```
else
{
  digitalWrite(led, LOW);
}

delay(200);
}
```

RESULT: Successfully demonstrated deploy smart street light system using LDR sensor.